

Santo Kabir Ahmed

Soil, Water and Environment Graduate

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Professional Profile

Postgraduate candidate with a proven track record in managing field surveys, laboratory analysis, and digital storytelling for environmental advocacy. Project Lead for a National Geographic Society-funded river pollution initiative, focusing on bridging the gap between high-resolution scientific data and community-led climate resilience. Eager to leverage strong background in analytical chemistry, geospatial analysis and extensive field-work experience to secure a scholarship for advanced academic research.

Education

- 2022 – 2026 **Bachelor of Science in Soil, Water and Environment**, *University of Dhaka*, Dhaka, CGPA: 3.82
Core focus on soil chemistry, ecology, and nature-based solutions.
- 2026 – **Master of Science in Soil, Water and Environment**, *University of Dhaka*, Dhaka
Current Core focus on Environment

Projects & Professional Experience

- June 2026 – **Research Assistant**, *Department of Soil Science*, Bangladesh Agricultural University
Current
 - Learned to operate various laboratory devices (spectrophotometer, multiparameter etc.).
 - Learned and utilized various wet lab analytical methods for soil and water sample analysis (Kjeldahl Method for Nitrogen, Wet-oxidation method for Carbon etc.).
 - Learned and participated in soil, water and gas (using gas chambers) sample collection from soil and water bodies.
 - Applied R programming for data analysis and visualization of soil, crop and water parameters.
 - Applied GIS for geospatial data analysis.
- Dec 2025 – **Project Lead**, *National Geographic Society*, Dhaleshwari River Pollution Awareness
May 2026
 - Spearheaded a National Geographic Society SEED-funded initiative to assess the impact of industrial discharge on 100+ downstream farming families.
 - Managed a dedicated group of over 20 volunteers.
 - Managed and conducted a comprehensive community survey of 56 residents to identify health correlations with river water usage.
 - Collected soil and water samples for assessment of the extent of industrial pollution.
 - Conducted awareness campaigns and an art competition across two local schools.
 - Created posters and an informative booklet for widespread knowledge dissemination.
- July 2025 – **Freshwater and Community Conservation Extern**, *National Geographic Society*, *The Nature Conservancy*, & Extern, Remote
Sep 2025
 - Developed the "Two Rivers, One Poison" ArcGIS StoryMap, integrating geospatial data with human-interest narratives to visualize the footprint of industrial pollution.
 - Conducted in-person field survey of the site of interest and interviews of local people for first-hand reliable data.
 - Collected, synthesized, and visualized a wide range of data using GIS software to enhance public understanding of environmental impacts.
 - Synthesized complex GIS layers into accessible, high-impact digital content.

- 2023 – 2024 **Undergraduate Researcher**, *Department of Soil, Water and Environment*, University of Dhaka
- Assisted in the analysis of food samples for heavy metal presence using standard laboratory protocols.
 - Maintained detailed, accurate documentation of experimental procedures and results for ongoing research projects.
 - Took a major part in writing and editing the manuscript for publication.

Publications

- 2024 Chamon, A. S., Parash, M. A. H., Fahad, J. I., Hassan, S. M. N., **Ahmed, S. K.**, Mushrat, M., Islam, N., Hasan, T., Atiya, Z., & Mondol, M. N. (2024). Heavy metals in dates (*Phoenix dactylifera* L.) collected from Medina and Dhaka City markets, and assessment of human health risk. *Environmental Systems Research*, 13, 27. (Q1 Journal)
<https://doi.org/10.1186/s40068-024-00354-7>
Key Contribution: Sample collection, laboratory analysis, and manuscript writing.

Technical Skills & Tools

Data Science & Software	R Programming, Python Programming (Advanced, University of Helsinki Certified), Microsoft Excel
Geospatial Analysis	QGIS, ArcGIS Online, ArcGIS Pro
Analytical Chemistry	Laboratory analysis for heavy metals (Cr, Pb, Cd) using Atomic Absorption Spectroscopy (AAS), Organic Carbon and different Nitrogen species analysis in Soil and Water
Environmental Analysis	Water and Soil Greenhouse Gas Sampling, Soil chemistry, Field sample collection & preservation

Language Skills

Bengali	Native Speaker
English	IELTS 8.00 (Listening: 9.0, Reading: 9.0, Writing: 7.5, Speaking: 6.5)

Volunteer, Community Leadership & Awards

- 13 Feb 2025 **2nd Runner-Up**, *Youth for Earth Case Solving Competition*, organized by the Ministry of Youth and Sports and University of Dhaka (Team: Trinnovia).
- Jan 2025 - Dec 2025 Student Ambassador and Communications Team member, *Nature Positive Universities (NPU)*.
- 2025 Volunteer, Clean Campus programs organized by *Dhaka University Environment Society (DUES)*.

References

Prof. Dr. Afrose Sultana Chamon
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 University of Dhaka
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Prof. Dr. M. M. R. Jahangir
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 Bangladesh Agricultural University (BAU)
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